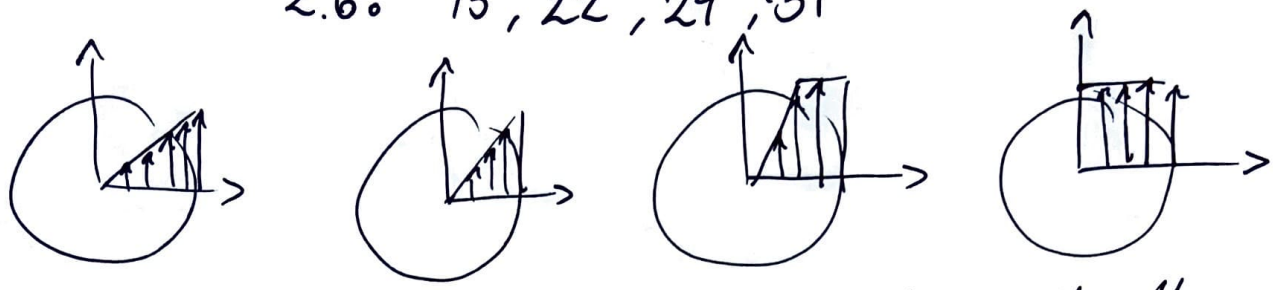
Övning 6 - Elastisk plastisk vridning

2.6: 15, 22, 29, 31



$M_v \leq M_s$ $M_v = M_s$ $M_s < M_v < M_f$ $M_v = M_f$

Vridande moment

$$M_v = 2\pi \int_a^b \tau(r) r^2 dr$$

Vridning

$$\Theta = \int_a^b \vartheta \, dx \qquad \vartheta = \frac{M}{GK}$$

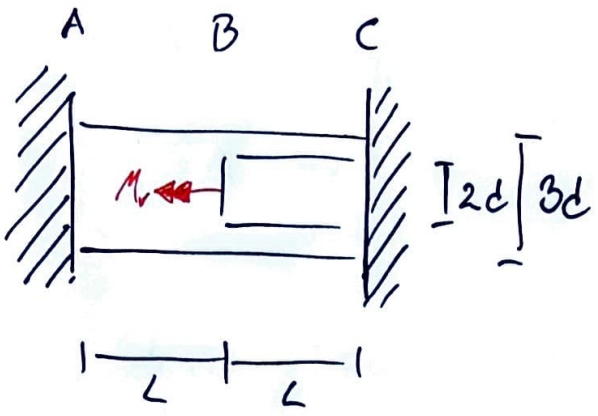
$$K = \frac{\pi}{32}(D^4 - d^4) \qquad [FS: 6.77]$$

d: inre diameter
D: yttre diameter

Skjuvspänning

$$\tau_{max} = \frac{M_v}{W_v} \qquad W_v = \frac{\pi}{16D}(D^4 - d^4) \qquad [FS: 6.78]$$

2.6.15

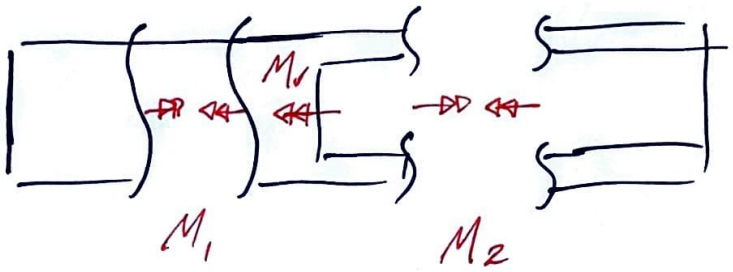


Elastiskt idealplastiskt

G, τ_s

Sökt: M_{vs} som ger plasticering

① Snittning



$$\circlearrowright: M_2 - M_1 - M_v = 0 \quad (1)$$

1 ekv
2 okända

② Deformationssamband

$$\theta_{tot} = \theta_1 + \theta_2 = 0 \quad (2)$$

③ Förvridning

$$\theta_i = \frac{M_i L}{G K_i}$$

$$K_1 = \frac{\pi}{32}(3d)^4 = \frac{81\pi d^4}{32}$$

$$K_2 = \frac{\pi}{32}\left((3d)^4 - (2d)^4\right) = \frac{65\pi d^4}{32}$$

$$\theta_1 = \frac{M_1 L}{G}\,\frac{32}{81\pi d^4}$$

$$\theta_2 = \frac{M_2 L}{G}\,\frac{32}{65\pi d^4}$$

ekv 2

$$\Rightarrow \quad \frac{M_1}{81} = -\frac{M_2}{65} \quad (3)$$

2.6.15 del 2

$$\begin{cases} M_2 - M_1 = M_v \\ 65M_1 = -81M_2 \end{cases} \Rightarrow \begin{cases} M_1 = -\frac{81M_v}{146} \\ M_2 = \frac{65M_v}{146} \end{cases}$$

④ Skjuvspänning

τ_{max} i de två delarna. Vilken plasticerar först?

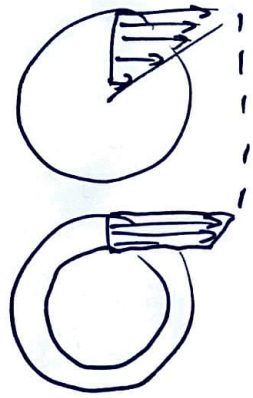
$$\tau_{max} = \frac{M_v}{W_v}$$

$$W_{v1} = \frac{\pi(3d)^4}{16 \cdot 3d} = \frac{27\pi d^3}{16}$$

$$W_{v2} = \frac{\pi((3d)^4 - (2d)^4)}{16 \cdot 3d} = \frac{65\pi d^3}{48}$$

$$\Rightarrow \tau_{max,1} = \frac{M_1}{W_1} = -\frac{24M_v}{73\pi d^3}$$

$$\tau_{max2} = \frac{M_2}{W_2} = \frac{24M_v}{73\pi d^3}$$



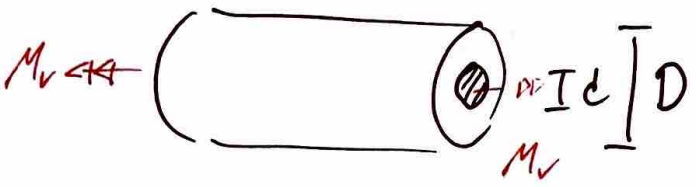
$|\tau_{max,1}| = |\tau_{max2}|$ Plasticerar samtidigt!

$$\tau_s = \frac{24M_{vs}}{73\pi d^3} \Rightarrow \boxed{M_{vs} = \frac{73}{24}\tau_s \pi d^3}$$

Enhets kontroll

$$\left[\frac{N}{m^2} m^3\right] = [Nm]$$

2.6.22



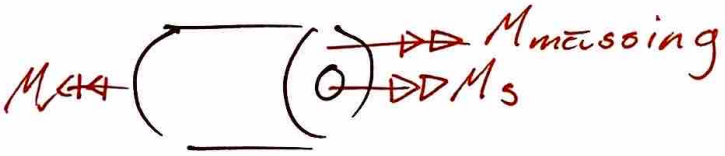
$G_{mässing} = 35\ GPa$

$G_{stål} = 80\ GPa$

$d = 30\ mm$

Sökt: Hur stor måste D vara för att mässingsröret och stålkärnan tar upp samma vridmoment?

① Snitta

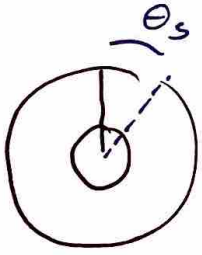


Jämvikt

$\Rightarrow: M_s + M_m = M_v$

2 okända
1 ekvation

② Deformationssamband



$$\theta_i = \frac{M_i L}{G_i K_i}$$

$$K_s = \frac{\pi d^4}{32}$$

$$K_m = \frac{\pi}{32}(D^4 - d^4)$$

$$\theta_s = \frac{M_s L}{G_s K_s} = \frac{M_s 32 L}{G_s \pi d^4}$$

$$\theta_m = \frac{M_m L}{G_m K_m} = \frac{M_m 32 L}{G_m \pi (D^4 - d^4)}$$

$$\theta_s = \theta_m \Rightarrow \frac{M_s}{G_s d^4} = \frac{M_m}{G_m (D^4 - d^4)} \quad (M_s = M_m)$$

$$\frac{1}{G_s d^4} = \frac{1}{G_m (D^4 - d^4)} \Rightarrow D^4 = \frac{d^4 (G_m + G_s)}{G_m} \Rightarrow D = 40.4\ mm$$

2.6.22 del 2

Sökt: Kvoten $\tau_{max,s} / \tau_{max,m}$

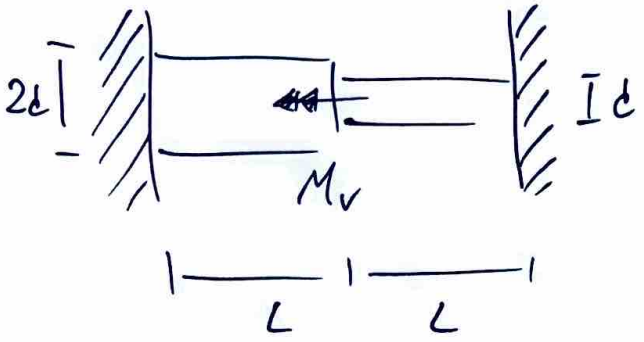
$$\tau_{max} = \frac{M_v}{W_v} \Rightarrow \frac{\tau_{max\,s}}{\tau_{max,m}} = \frac{M_s}{W_{vs}} \frac{W_{vm}}{M_v} =$$

$$= \frac{W_{vm}}{W_{vs}} = \left\{ \begin{array}{l} W_{vm} = \frac{\pi}{16D}(D^4 - d^4) \\ W_{vs} = \frac{\pi}{16} d^3 \end{array} \right\} =$$

$$= \frac{\frac{\pi}{16D}(D^4 - d^4)}{\frac{\pi}{16} d^3} = \frac{D^4 - d^4}{d^3 D} = 1.7$$

Skjuvspänningen är 70% högre i stål-kärnan än mässingen

2.6.29

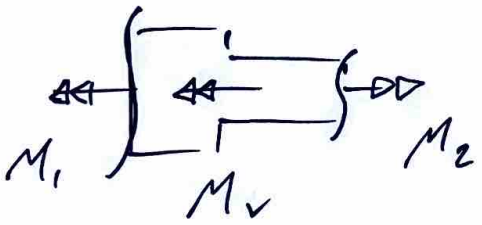


Givet: τ_s, G

Sökt: $\frac{M_f}{M_s}$ - Totalplasticering
- Begynnande plasticering

① Snitta

Jämvikt



$\rightarrow$: $M_2 = M_1 + M_v$

② Deformationssamband

$$\theta_{tot} = \theta_1 + \theta_2 = 0$$

$$K_1 = \frac{\pi d^4}{2} \qquad K_2 = \frac{\pi d^4}{32}$$

$$\Rightarrow \quad \theta_1 = \frac{M_1 L}{G} \frac{2}{\pi d^4} \qquad \theta_2 = \frac{M_2 L}{G} \frac{32}{\pi d^4}$$

$$\Rightarrow \quad 2M_1 = -32M_2 \Rightarrow M_1 = -16M_2$$

$$\begin{cases} M_2 = M_1 + M_v \\ M_1 = -16M_2 \end{cases} \Rightarrow \begin{cases} M_1 = -\frac{16}{17} M_v \\ M_2 = \frac{1}{17} M_v \end{cases}$$

③ Begynnande plasticering

$$\tau_{max} = \frac{M_v}{W_v} \qquad W_{v1} = \frac{\pi d^3}{2} \qquad W_{v2} = \frac{\pi d^3}{16}$$

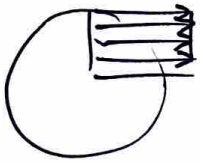
$$\tau_{max1} = -\frac{16 M_v}{17}\frac{2}{\pi d^3} = -\frac{32 M_v}{17\pi d^3}$$

$$\tau_{max2} = \frac{M_v}{17}\frac{16}{\pi d^3} = \frac{16 M_v}{17\pi d^3}$$

Tvärsnitt 1 plasticerar först

$$\tau_s = \frac{32 M_{vs}}{17\pi d^3} \Rightarrow M_{vs} = \frac{17}{32}\tau_s \pi d^3$$

④ Full plasticering



$\tau(r) = \tau_s$ för alla r

Jämvikt gäller fortfarande

$$M_1 + M_v = M_2 \Rightarrow M_{f,1} + M_f = M_{f,2}$$

$$M_v = 2\pi \int_0^d \tau(r) r^2 dr \Rightarrow M_{f,i} = 2\pi \int_0^d \tau_s r^2 dr$$

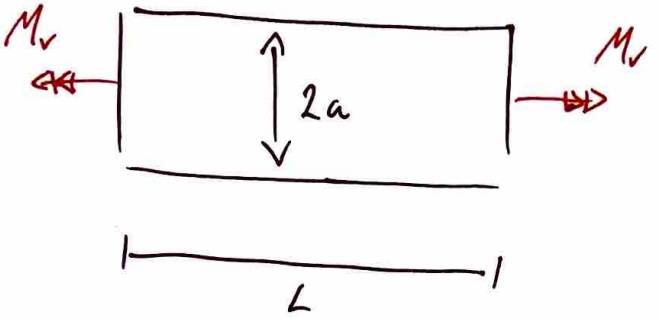
$$M_f = M_{f,2} - M_{f,1} = 2\pi \int_0^{d/2} \tau_s r^2 dr + 2\pi \int_0^d (-\tau_s) r^2 dr$$

$$M_f = 2\pi\tau_s \left(\left[\frac{r^3}{3}\right]_0^{d/2} + \left[\frac{r^3}{3}\right]_0^{d} \right) = \frac{3}{4}\pi\tau_s d^3$$

$$\frac{M_f}{M_s} = \frac{\frac{3}{4}\pi\tau_s d^3}{\frac{17}{32}\pi\tau_s d^3} = \frac{24}{17} \simeq 1{,}4$$

2.6 : 31

Givet: a, L, G, τ_s



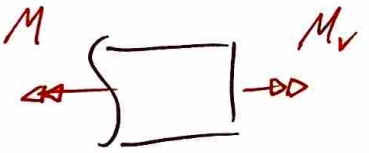
Stången belastas med $M_v = 1.2 M_s$ där M_s är vridning för begynnande plasticitet

Sökt: a) Plasticerings djupet. r_s

b) Återfjädring och kvarvarande deformation efter avlastning θ_R

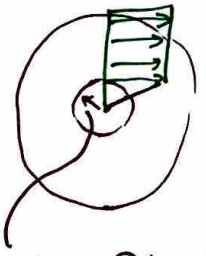
c) Restspänningen $\tau_{rest}(r)$

① Snittning



$M = M_v$ samma belastning genom hela stången

② Uttryck för M_v



r_s - Plasticeringsdjup

Över plasticerings djupet $\tau(r)$ konstant

Under plasticeringsdjupet $\tau(r)$ linjärt

Elastiskt område

$$\tau(r) = \tau_s \frac{r}{r_s} \quad | r < r_s |$$

Plastiskt område

$$\tau(r) = \tau_s \quad | r > r_s |$$

$$M_v = 2\pi \int_0^a \tau(r) r^2 dr$$

2.6:31 del 2

$$M_v = 2\pi \int_0^{r_s} \tau_0 \frac{r}{r_s} r^2 dr + 2\pi \int_{r_s}^{a} \tau_s r^2 dr$$

$$\Rightarrow M_v = 2\pi \tau_s \int_0^{r_s} \frac{r^3}{r_s} dr + 2\pi \tau_s \int_{r_s}^{a} r^2 dr$$

$$M_v = 2\pi \tau_s \left(\left[\frac{r^4}{4 r_s} \right]_0^{r_s} + \left[\frac{r^3}{3} \right]_{r_s}^{a} \right)$$

$$M_v = \frac{2}{3}\pi \tau_s \left(a^3 - \frac{r_s^3}{4} \right) \Rightarrow r_s = \sqrt[3]{4\left(a^3 - \frac{3}{2}\frac{M_v}{\pi \tau_s} \right)}$$

③ Bestäm M_v

$M_v = 1.2 M_s \Rightarrow M_s$? Initiell plasticering

$r_s = a$

$$M_s = \frac{2}{3}\pi \tau_s \left(a^3 - \frac{a^3}{4} \right) = \frac{\pi \tau_s a^3}{2}$$

alternativt

$$\tau_{max} = \tau_s = \frac{M_s}{W_v} \Rightarrow \tau_s = \frac{2M_s}{\pi a^3} \Rightarrow M_s = \frac{\pi \tau_s a^3}{2}$$

$$M_v = 1.2 M_s = \frac{1.2 \pi \tau_s a^3}{2}$$

$$r_s = \sqrt[3]{4\left(a^3 - \frac{3 \cdot 1.2}{4} \frac{\tau_s a^3 \pi}{\pi \tau_s} \right)} = \sqrt[3]{0.4}\, a$$

$$\boxed{r_s = 0.737a}$$

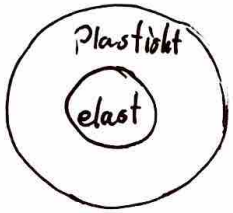
[Går att lösa med FS. 6.85]

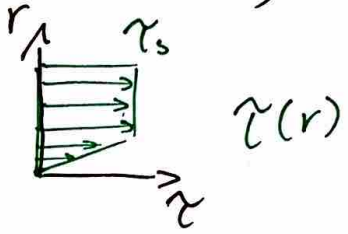
2.6.31 del 3

b) återfjädring och deformation

För elastisk förvridning gäller

$$\Theta = \frac{2M_v L}{G\pi a^4} \qquad \left(\frac{M_v L}{GK} \quad K = \frac{\pi}{2}a^4\right)$$





$$\Theta = \frac{2M_v^{elast} L}{G\pi r_s^4} \qquad \text{Moment i elastisk del}$$

$$M_{v,elast} = 2\pi \int_0^{r_s} \tau_s \frac{r^3}{r_s} dr = \frac{1}{2}\pi \tau_s r_s^3$$

Totala förvridningen vid pålast

$$\Theta = \frac{2M_{v,elast} L}{G\pi r_s^4} = \frac{2L}{G\pi r_s^4}\frac{1}{2}\pi \tau_s r_s = \frac{\tau_s L}{G r_s}$$

Avlastning sker med $M_v = -1.2 M_s$

$$\Theta_{avlast} = -\frac{2.4 M_s L}{G\pi a^4} = \left\{M_s = \frac{\tau_s \pi a^3}{2}\right\} = -1.2\frac{\tau_s L}{Ga}$$

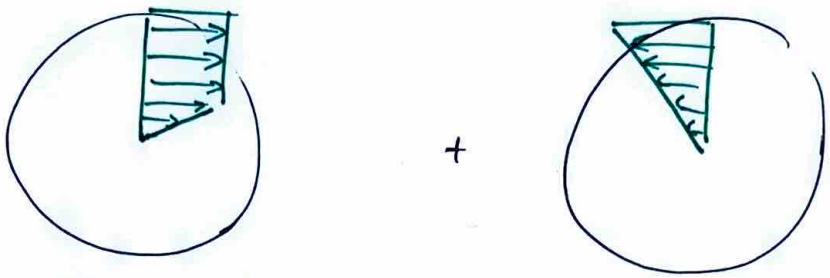
$$\Theta_R = \Theta + \Theta_{avlast} = \frac{\tau_s L}{G r_s} - 1.2\frac{\tau_s L}{Ga} = \left\{r_s = 0.737a\right\} =$$

$$\Rightarrow \boxed{\Theta_R = 0.157\frac{\tau_s L}{Ga}}$$

2.6:31 del 4

c) Restspänningar

Pålastning + Avlastning



$M_v = 1.2 M_s$ $\qquad$ $M_v = -1.2 M_s$

$$\tau(r) = \begin{cases} \tau_s \frac{r}{r_s} & r < r_s \\ \tau_s & r > r_s \end{cases} \qquad \tau(r) = \tau_{max,a} \frac{r}{a}$$

$$\tau_{max,a} = \frac{-1.2 M_s}{\frac{\pi}{2} a^3} = \left(\frac{M_v}{W_v} \right) = -1.2 \tau_s$$

Avlastning

$$\tau(r) = -1.2 \frac{\tau_s r}{a}$$

Superpositionering

Restskjuvspänning

$$\tau(r)_R = \begin{cases} \tau_s \frac{r}{r_s} - 1.2 \tau_s \frac{r}{a} & r < r_s \\ \tau_s - 1.2 \tau_s \frac{r}{a} & r > r_s \end{cases} = \begin{cases} 0.157 \tau_s \frac{r}{a} & r < r_s \\ (1 - 1.2 \frac{r}{a}) \tau_s & r > r_s \end{cases}$$

Illustrerat

